EMPLOYER-SET PROJECT PRACTICE

ESP Practice Lab

Practice for the Packet Tracer parts of the ESP — finding and fixing security defects and securing a network extension — plus the discipline that wins the marks. Original practice material, not the live assessment.

This is original practice material built around the task types. It is not the live assessment and uses a fictional scenario.

How marks are won in the practical tasks

- **Test systematically.** For every requirement: a planned test, the **expected** result, the **actual** result, the **action** taken — and a **re-test** to prove the fix. A list of visual checks with no re-test stays in the lowest band.
- **Use a range of test methods:** visual inspection, the command line (ping, ipconfig), and the device console.
- **Apply precise, secure configurations** (checklist below). Fully addressing the vulnerabilities moves the solution into the top band.
- **Be consistent:** follow the existing naming conventions and a logical IP addressing scheme — this earns the organisation marks.
- **Reason, don't list.** In design and evaluation, justify choices and weigh trade-offs. In the evaluation, review **your own solution**, not a re-run of your plan.

Test-log structure

Requirement / test	Expected	Actual	Action & re-test
PC VLAN membership	VLAN 10	VLAN 1 (default)	switchport access vlan 10 — re-tested: show vlan brief confirms
Wireless encryption	WPA2 (AES)	WEP	reconfigured to WPA2-PSK — re-tested: connects only with WPA2

Secure-config conventions checklist

Area	Secure choice
Wireless	WPA2-PSK (AES); hidden, non-descriptive SSID
Switch ports	disable unused ports; port security / MAC filtering
Management	HTTPS & SSH only — never HTTP/Telnet
Accounts	named, least-privilege; strong hashed passwords (no defaults)
Addressing	logical scheme; right-sized DHCP scopes; no overlaps
Segmentation	VLANs per function; isolate public/guest devices
Perimeter	enable the firewall; deny unsolicited inbound
Backups	secured account; tested recovery

Three kinds of risk to cover

A strong security plan covers all three — don't stop at the technical ones:

- **Physical:** locks, cable/switch access, device theft, shoulder-surfing, tailgating.

- **Logical / software:** weak encryption, open ports/services, unpatched systems, malicious QR codes/links.
- **User-based:** phishing & social engineering, unlocked devices, weak passwords, poor training.